

Project Report

Title: **HEALTHCARE APPOINTMENT NO-SHOW PREDICTION**

Abstract

This project focuses on predicting whether a patient will miss a scheduled healthcare appointment using machine learning and visual analytics. The dataset contained 110,521 appointment records. Exploratory data analysis was performed to identify factors associated with no-shows, followed by data preprocessing, feature engineering, and model development. Logistic Regression, Decision Tree, and Random Forest models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC. Since the dataset was imbalanced and the primary objective was to identify potential no-shows, recall was given particular importance. Random Forest was selected as the final model, achieving 82.81% recall, 44.34% F1-score, and 72.30% ROC-AUC. A Power BI dashboard was developed to present appointment patterns, risk categories, and model performance.

1. Introduction

Missed healthcare appointments, or no-shows, can lead to inefficient use of medical resources, longer waiting times, and reduced healthcare accessibility. Predicting patients who are likely to miss appointments can help healthcare providers take preventive measures such as targeted reminders and appointment confirmations.

The objective of this project is to develop a machine learning-based Appointment No-Show Prediction System that identifies appointments at higher risk of being missed. The project combines Python-based data analysis and machine learning with Power BI visualization to provide both predictive insights and an interactive dashboard.

2. Tools Used

- Python – Data cleaning, preprocessing, exploratory data analysis and machine learning
- Pandas & NumPy – Data manipulation and numerical processing
- Scikit-learn – Model development and evaluation
- Jupyter Notebook – Data analysis and model implementation
- Power BI – Interactive dashboard and visualization

3. Steps Involved in Building the Project

1. Data Collection and Understanding:

The appointment dataset was examined to understand its structure, variables, target variable, and data distribution. The final dataset contained 110,521 records and 18 original variables, with no missing or duplicate records.

2. Data Preprocessing:

The data was cleaned and prepared for machine learning. Relevant variables were selected,

categorical variables were encoded, numerical features were processed, and the target variable was created with 0 representing attended appointments and 1 representing no-shows.

3. Exploratory Data Analysis:

Appointment characteristics and relationships with no-show behaviour were analysed. Important variables included waiting days, SMS reminders, scholarship status, and age. Waiting days showed the strongest relationship among the examined variables.

4. Machine Learning:

Three classification algorithms were developed and compared:

- Logistic Regression
- Decision Tree
- Random Forest

The models were evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

5. Model Evaluation and Selection:

Although Logistic Regression achieved higher accuracy (79.5%), its no-show recall was only 1.5%. Decision Tree achieved 80.0% recall, while Random Forest achieved the highest recall of 82.81%, along with the highest F1-score (44.34%) and ROC-AUC (72.30%). Therefore, Random Forest was selected as the final model because detecting actual no-shows was more important than maximizing overall accuracy.

6. Risk Analysis:

Predictions were converted into risk categories. The observed no-show rates were 3.55% for Low Risk, 20.78% for Medium Risk, and 36.89% for High Risk, demonstrating that the risk classification effectively differentiated appointments according to their likelihood of no-show.

7. Power BI Dashboard:

A three-page Power BI dashboard was developed:

- Page 1 – Appointment Overview: Overall appointment and attendance analysis.
- Page 2 – Risk Analysis: Risk categories and no-show patterns.
- Page 3 – Model Performance: Model comparison, Random Forest performance metrics, and confusion matrix.

4. Conclusion

The project successfully developed a machine learning-based approach for identifying appointments at risk of no-show. The analysis demonstrated that factors such as waiting time and appointment reminders can provide useful predictive information. Among the evaluated models, Random Forest was selected as the final model because of its superior recall, F1-score, and ROC-AUC. The Power BI dashboard presents the results in an accessible form and can support healthcare staff in prioritizing high-risk appointments for additional reminders or follow-up. Overall, the project demonstrates how data science and predictive analytics can contribute to more efficient healthcare appointment management.